3.3. Representations of direct sums of matrix algebras

In this section we consider representations of algebras $A = \bigoplus_i \mathrm{Mat}_{d_i}(k)$ for any field k.

Theorem 3.3.1. *Let $A = \bigoplus_{i=1}^r \mathrm{Mat}_{d_i}(k)$. Then the irreducible representations of A are $V_1 = k^{d_1}, \dots, V_r = k^{d_r}$, and any finite dimensional representation of A is a direct sum of copies of $V_1, \dots, V_r$.*

In order to prove Theorem 3.3.1, we shall need the notion of a dual representation.

Definition 3.3.2 (Dual representation)**.** Let V be a representation of any algebra A. Then the **dual representation** V^* is the representation of the opposite algebra A^{op} (or, equivalently, right A-module) with the action

$$(f \cdot a)(v) := f(av).$$

Proof of Theorem 3.3.1. First, the given representations are clearly irreducible, since for any $v \neq 0, w \in V_i$, there exists $a \in A$ such that $av = w$. Next, let X be an n-dimensional representation of A. Then, X^* is an n-dimensional representation of A^{op}. But $(\mathrm{Mat}_{d_i}(k))^{\mathrm{op}} \cong \mathrm{Mat}_{d_i}(k)$ with isomorphism $\varphi(X) = X^T$, as $(BC)^T = C^T B^T$. Thus, $A \cong A^{\mathrm{op}}$ and X^* may be viewed as an n-dimensional representation of A. Define

$$\phi : \underbrace{A \oplus \cdots \oplus A}_{n \text{ copies}} \longrightarrow X^*$$

by

$$\phi(a_1, \dots, a_n) = a_1 y_1 + \cdots + a_n y_n$$

where $\{y_i\}$ is a basis of X^*. The map ϕ is clearly surjective, as $k \subset A$. Thus, the dual map $\phi^* : X \longrightarrow A^{n*}$ is injective. But $A^{n*} \cong A^n$ as representations of A (check it!). Hence, $\mathrm{Im}\,\phi^* \cong X$ is a subrepresentation of A^n. Next, $\mathrm{Mat}_{d_i}(k) = d_i V_i$, so $A = \bigoplus_{i=1}^r d_i V_i$, $A^n = \bigoplus_{i=1}^r n d_i V_i$, as a representation of A. Hence by Proposition 3.1.4, $X = \bigoplus_{i=1}^r m_i V_i$, as desired. $\square$